EXPLORATORY DATA ANALYSIS (EDA) PROJECT

Analysis of Student Academic Performance Dataset

Project Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on a student academic performance dataset. The analysis uses statistical summaries and visualizations to understand the data, identify patterns and trends, study relationships between variables, and identify factors that are associated with examination performance.

Expected Outcome

The project develops analytical thinking and practical data exploration skills by demonstrating how raw data can be examined, summarized, visualized, and interpreted to produce meaningful insights.

1. Introduction

Exploratory Data Analysis is the process of investigating a dataset before applying advanced statistical or machine learning techniques. EDA helps identify the structure of data, missing or unusual observations, distributions, relationships, and important variables. In this project, student academic performance is used as an example because exam scores can be explored in relation to study habits, attendance, assignments, sleep, and internet usage.

2. Dataset Description

For this project, an illustrative dataset containing 200 student records was prepared. Each record represents one student and contains academic and lifestyle-related variables. The dataset is designed for demonstrating the EDA workflow and should be treated as an academic project dataset rather than as an official institutional survey.

Variable	Type	Description
Study Hours	Numerical	Average study hours per day
Attendance (%)	Numerical	Percentage of classes attended
Assignment Score (%)	Numerical	Average assignment performance
Sleep Hours	Numerical	Average sleep hours per day
Internet Usage (hrs)	Numerical	Average daily internet usage
Exam Score (%)	Numerical	Final examination score

3. EDA Methodology

Step 1 – Data Understanding: Identify rows, columns, variable types, and the purpose of each variable.

Step 2 – Data Cleaning: Check missing values, duplicate records, inconsistent values, and potential outliers.

Step 3 – Statistical Summary: Calculate count, mean, median, standard deviation, minimum, maximum, and quartiles.

Step 4 – Univariate Analysis: Study individual variables using distributions and summary statistics.

Step 5 – Bivariate Analysis: Examine relationships between pairs of variables using scatter plots and correlation.

Step 6 – Interpretation: Identify important patterns and convert them into clear analytical findings.

4. Data Quality Checks

The dataset contains 200 records and 6 variables. In this prepared dataset, no missing values or duplicate records were intentionally included. Numerical ranges were reviewed to ensure that attendance and scores remain within logical percentage limits. In a real-world project, these checks should be followed by appropriate treatment of missing values, duplicates, invalid records, and outliers.

5. Statistical Summary

Variable	Mean	Median	Std. Dev.	Min	Max
Study Hours	5.92	5.99	1.66	1.28	10.00

Attendance (%)	78.79	78.79	9.53	50.00	100.00
Assignment Score (%)	73.90	74.07	11.74	45.34	100.00
Sleep Hours	7.01	7.02	1.02	4.30	9.63
Internet Usage (hrs)	4.20	4.20	1.42	1.00	7.79
Exam Score (%)	90.97	91.16	7.40	66.81	100.00

The mean and median values provide an initial understanding of the central tendency, while standard deviation indicates how widely observations vary around the average. Comparing the minimum and maximum values also helps detect unusually broad ranges.

6. Univariate Analysis

Univariate analysis examines one variable at a time. The exam-score distribution helps identify whether student performance is concentrated around a particular range or spread across a wider range.

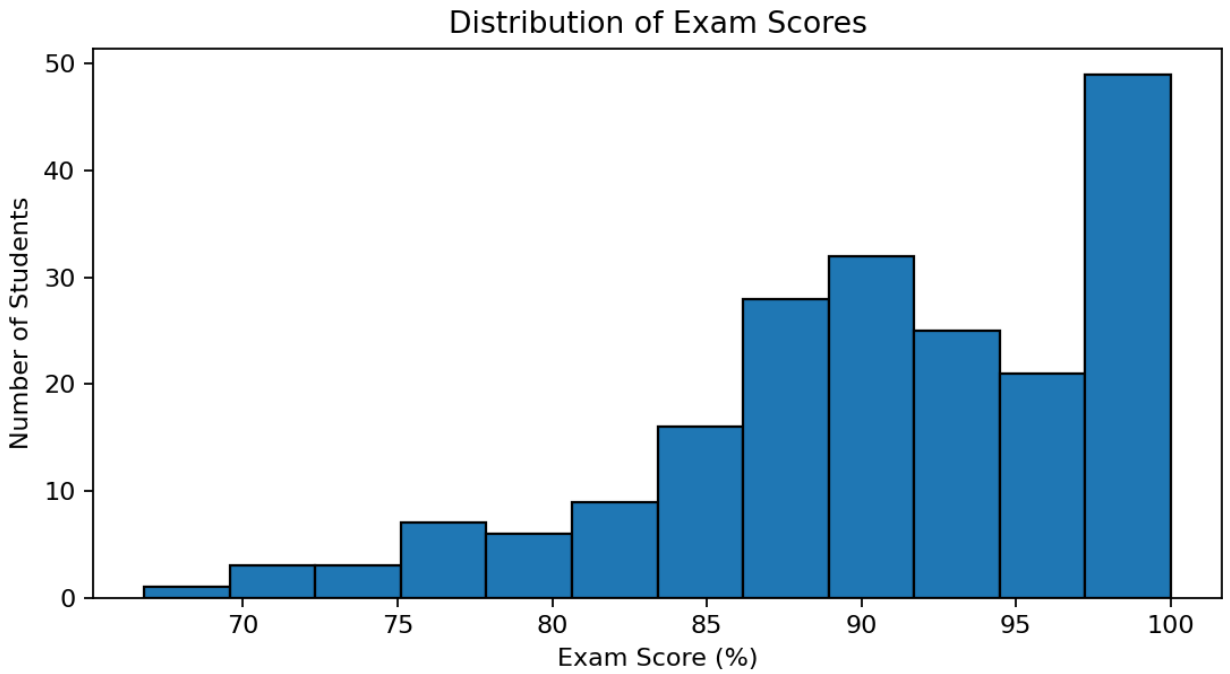


Figure 1: Distribution of examination scores.

The distribution provides a visual overview of performance. A histogram is useful for identifying concentration, spread, possible skewness, and unusual observations. Similar plots can be created for attendance, study hours, assignment scores, and sleep hours.

7. Bivariate Analysis

Bivariate analysis studies the relationship between two variables. In this project, study hours are compared with exam scores because study time is a plausible academic factor.

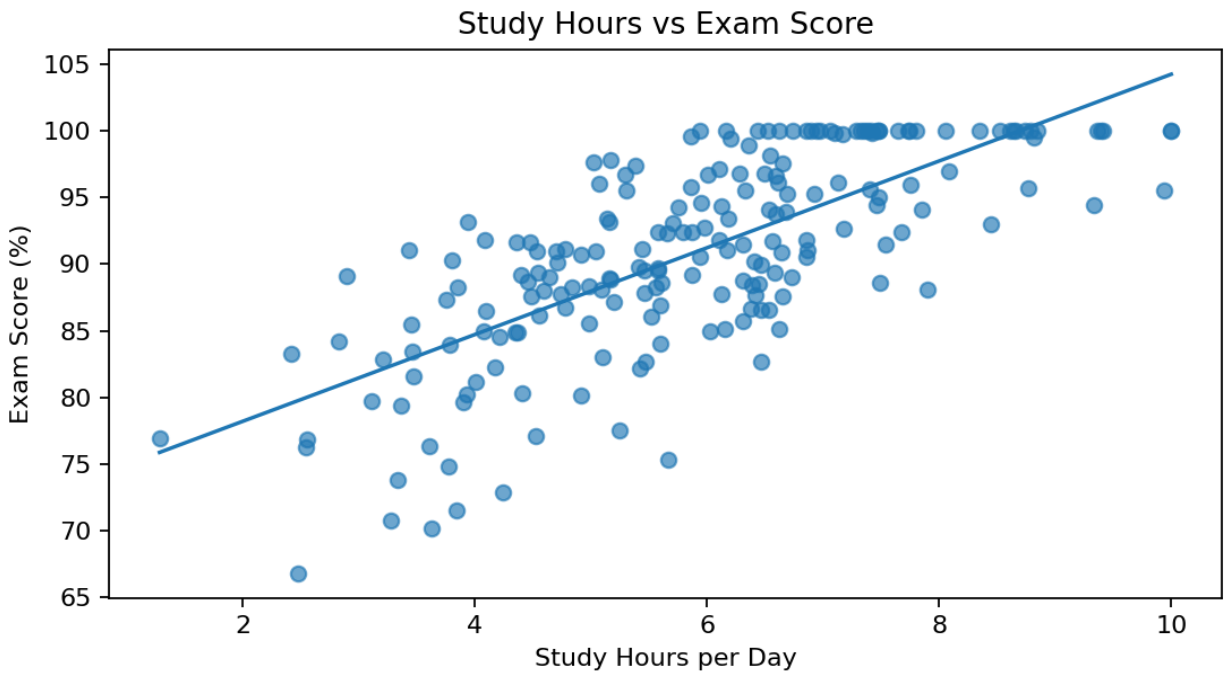


Figure 2: Relationship between study hours and exam score.

The scatter plot shows the overall direction and strength of the relationship. The fitted line summarizes the linear trend, but correlation does not prove that increased study time alone causes higher scores. Other variables may also contribute.

8. Correlation Analysis

Correlation measures the strength and direction of a linear relationship between numerical variables. Values closer to +1 indicate stronger positive association, values closer to -1 indicate stronger negative association, and values near 0 indicate weak linear association.

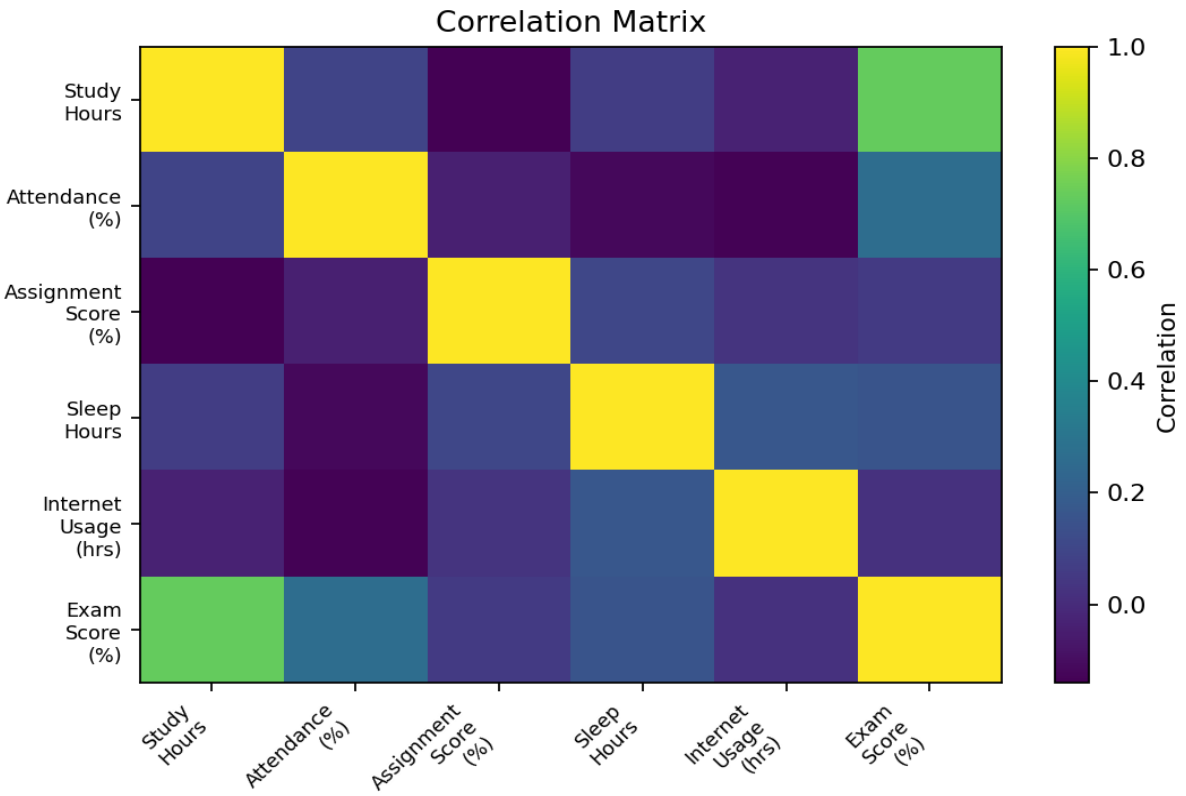


Figure 3: Correlation matrix of the numerical variables.

Variable	Correlation with Exam Score
Study Hours	0.730
Attendance (%)	0.265
Sleep Hours	0.156
Assignment Score (%)	0.052
Internet Usage (hrs)	0.020

In this illustrative dataset, **Study Hours** has the strongest positive linear correlation with exam score among the listed variables. This indicates an association in the dataset, not a causal relationship. Correlation should therefore be interpreted together with domain knowledge and, where appropriate, further statistical or predictive analysis.

9. Key Insights

• Exam scores show measurable variation across students, making distribution analysis useful for understanding overall performance.

• Study hours show a positive relationship with exam scores in the illustrative dataset.

• Attendance and assignment performance can also be examined as potential influencing factors because they have direct academic relevance.

• Correlation helps identify variables worth further investigation, but it should not be interpreted as proof of cause and effect.

• Visualization makes relationships and distributions easier to understand than relying only on numerical summaries.

10. Business/Academic Interpretation

From an analytical perspective, EDA can help educational institutions identify patterns that deserve further investigation. For example, if attendance, assignment performance, and study time consistently show relationships with examination results, these variables can be included in subsequent predictive modeling or student-support analysis. Any real institutional decision should use validated data, appropriate statistical testing, and ethical handling of student information.

11. Tools and Technologies

Python: Programming language used for data analysis.
Pandas: Data loading, cleaning, transformation, and statistical summaries.
NumPy: Numerical operations and dataset preparation.
Matplotlib: Data visualization and chart creation.
Jupyter Notebook: Interactive environment for conducting and documenting EDA.

12. Conclusion

This Exploratory Data Analysis project demonstrates a complete basic EDA workflow: understanding the dataset, checking data quality, generating statistical summaries, visualizing distributions, examining relationships, calculating correlations, and interpreting findings. The project develops practical analytical thinking by converting raw observations into structured insights. The same workflow can be applied to datasets from healthcare, finance, marketing, retail, education, and many other domains.

13. Future Scope

The project can be extended by collecting a larger real-world dataset, performing formal outlier and missing-value treatment, using hypothesis testing, applying feature selection, building regression or classification models, evaluating model performance, and developing an interactive dashboard using tools such as Power BI or Tableau.

14. References / Learning Resources

1. Python documentation and standard data-analysis libraries.
2. Pandas documentation for data manipulation and analysis.
3. Matplotlib documentation for visualization.
4. Standard introductory resources on Exploratory Data Analysis and descriptive statistics.